

moving objects or persons to be detected. The integrated I/Q signalling method also enables directions to be detected. The antenna design boasts a detection range of $72^\circ \times 36^\circ$ (field of view) and a typical range for the detection of people up to 15 metres. The IMD-1100 is well-suited for large-scale use, especially for pick-and-place production.

InnoSenT GmbH
www.innosent.de

Ultra-fast charging IC for smartphones

NuVolta Technologies, a provider of fast charging power ICs, has announced its second-generation charge pump fast-charging IC, NU2205, which is the first 100W charge pump fast-charging IC in the whole industry. NU2205 is a

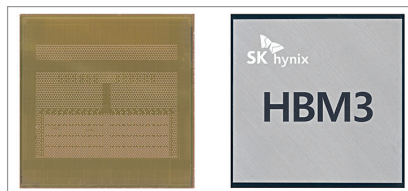


fast-charging IC for 2S battery with 4:2 charge pump fast-charging architecture for ultra-fast charging applications in flagship smartphones and other mobile devices. The innovative 2S battery fast-charging architecture enables the charging power from 60W to 120W, which can be further increased to 200W.

NuVolta Technologies
www.nuvoltatech.com

DRAM that can transmit 163 full-HD movies per second

SK hynix has successfully developed the High Bandwidth Memory 3 (HBM3), the world's best-performing DRAM. HBM3 is the fourth generation product by the company that is based on HBM technology with a combination of multiple



DRAM chips vertically connected. HBM3 can process up to 819GB per second, meaning that 163 FHD (full-HD) movies (5GB each) can be transmitted in a single second. This represents a 78% increase in the data-processing speed compared with its predecessor, the HBM2E. HBM3 is suitable for high-performance data centres as well as machine learning platforms to conduct climate change analysis and drug development.

SK hynix Inc.
www.skhynix.com

In-cabin continuous wireless charging solution

Spark Connected announces The Beast 2.0, a 15W automotive in-cabin wireless charging solution that meets the latest Qi requirements. The Beast 2.0 offers improved EMC performance, allowing customers an easy-to-implement

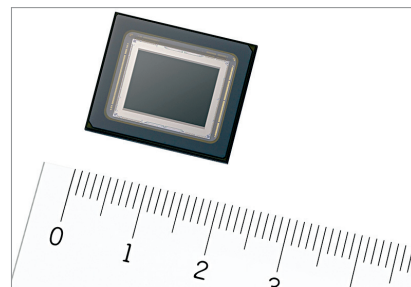


solution that meets strict OEM requirements. In addition to its impressive EMC capability, the Beast 2.0 also meets all the latest Qi requirements and provides class-leading safety features and a wide input supply range that provides continuous charging even during vehicle stop/start events. Foreign object detection is improved over other solutions in the market, increasing user safety while still providing a flexible charging experience.

Spark Connected
www.sparkconnected.com

Compact UV-sensitive industrial CMOS image sensor

Sony Semiconductor has announced commercialisation of the IMX487, a 2/3-type CMOS image sensor for industrial equipment, which is compatible with the UV wavelength and comes with



global shutter functionality with a high effective pixel count of about 8.13 megapixels. It delivers a high level of UV sensitivity and high-quality imaging with minimal noise thanks to components specialised for the UV wavelength and a unique light receiving unit structure. Additionally, the pixels capture UV light with high efficiency, resulting in high UV sensitivity as well as a compact form factor and high resolution.

Sony Semiconductor Solutions Corporation
www.sony-semicon.co.jp

Low-noise scientific camera measures the number of photons

Hamamatsu Photonics introduces a new scientific camera, the ORCA-Quest qCMOS.

It has incredibly low noise of 0.27 electrons RMS and a high pixel count of



9.4 megapixels. The ORCA-Quest reduces the photoelectric noise caused when light is converted into electrical signals, to a level below the signals generated by photons (particles of light), which are the minimum unit of light. This makes the ORCA-Quest the world's first camera to achieve 2D photon-number-resolving measurement, meaning that it accurately measures the number of photons to create an image. Its ability to identify the number of photons is optimal for applications in the astronomical research and life science fields.

Hamamatsu Photonics
www.hamamatsu.com